AI FOR GOOD SANDBOX HACKATHON · TECHNICAL REPORT

ARGOS Transit Agents

Policy-grounded coordination for container transit in Cameroon

Team leader: to be confirmed · Team members: to be confirmed · Contact: to be confirmed

Repository: github.com/bradymickael400-dotcom/Projet_de_fin_d_Etude · Verify public access before submission

Submission status: working draft. Operational claims below are tied to code or saved evidence; simulated outputs and unresolved legal questions are identified explicitly.

1. Introduction

A container moving from Douala toward an inland destination passes through several organizations and handovers. Its shipment declaration, seal state, location, and condition may be recorded in separate systems and may arrive at different times. During an incident, this makes it harder for responsible people to agree on what is known, what is uncertain, who should verify the next fact, and whether information can be shared.

ARGOS Transit Agents is a prototype for rehearsing those questions through staged, policy-aware scenarios. Four simulated roles—logistics operator, customs/port, data protection, and policy adviser—receive different facts and search separate partitions of a local knowledge base. A deterministic coordinator releases events in sequence, records evidence and uncertainty, and holds a proposed personal-location transfer for human review. The roles recommend checks; they do not impersonate public agencies, decide legal responsibility, or take physical action.

The existing ARGOS IoT tracking application is a supporting evidence channel. The policy scenario is the central submission contribution. Its cargo, people, and sensor events are synthetic. The prototype does not demonstrate customs deployment, field-validated AI, legal compliance, or certification.

2. Use case and problem addressed

The primary scenario, silent_container_v1, follows a fictional temperature-sensitive shipment from the port of Douala to an inland yard in Cameroon. The sequence deliberately introduces incomplete and conflicting information: an intact seal is reported at departure; a door event, temperature rise, and GPS loss occur later; a location-sharing request arrives without a stated purpose or retention period; a delayed device reading places the event in a yard where a broken seal is recorded; and an inspection is proposed while the fictional cold-chain deadline approaches.

The intended value is procedural: preserve which role saw which facts and passages, make missing evidence visible, permit reassessment when late evidence arrives, and route sensitive decisions to a person. ARGOS does not establish whether private telemetry is legally admissible evidence, determine which authority owns a handover, or replace customs, port, rail, road, or carrier systems.

3. Use-case requirements and current evidence

Requirement	Current implementation / evidence	Status and boundary
SRC · Provenance	Scenario events have IDs and SIMULATED origin. The source register records authority, jurisdiction, URL, verification status, access/licence note, language and role partitions for six policy references.	Implemented for the demonstration. The corpus is selective; not every item is a complete primary-text ingestion.
C · Collection and coordination	A deterministic coordinator releases six events in sequence. The separate IoT API has an authenticated telemetry path and validates device/box association and sequence.	Scenario replay runs without an external LLM. The policy replay does not ingest its synthetic events into operational IoT storage.
PP · Validation and preparation	Scenario facts are assigned to roles; local lexical retrieval returns passage paths and line ranges. IoT telemetry validation is a separate application function.	No claim of complete personal-data pseudonymisation or article-level completeness across the corpus.
M · Methods and agents	Four simulated policy roles; deterministic workflow and review gates; lexical document search; optional provider-assisted wording.	No legally validated agent model. External LLM availability and real-data suitability are not demonstrated.
P · Policy and decision	T3 location transfer is blocked pending human review; every stage has a pending human gate and a reviewable summary.	A software gate does not itself establish that a transfer is lawful or define an official decision-maker.
D / SINK · Delivery and people	Role-specific summaries, source references, uncertainty and review status are in the scenario trace. Human reviewers remain accountable.	English/French reports exist. Full bilingual product interface, accessibility and institutional operating procedure are not verified.

4. Architecture mapping: ITU-T Y.3172 and AI Readiness

The organizer asks for the Y.3172 pipeline and the AI Readiness framework. The mapping below shows where ARGOS components fit; it is not a certification. In Y.3172, C is the Collector. The 2025 ITU AI Ready report describes readiness dimensions and domain metrics; ARGOS has not calculated a readiness score. The report considers data, digital infrastructure, devices/usability, automation and access to AI, skills, innovation ecosystem, standards, and AI policy/implementation. We use the dimensions as questions for evidence, not as labels implying maturity.

Pipeline node	ARGOS component and data flow	Readiness / policy questions and evidence boundary
SRC · Sources	Synthetic scenario facts; source register and selected Cameroon/WTO material. Optional ESP32/telemetry evidence is separate.	Provenance and quality are explicit for the scenario. Field representativeness, sensor calibration, licensing and full legal-source coverage remain open.
C · Collector	Deterministic scenario coordinator; separately, authenticated IoT collection API.	Collection reliability, connectivity, device availability and organizational coordination matter. Offline recovery and regional infrastructure are not demonstrated by a replay.
PP · Pre-processing	Event sequence and role assignment; lexical retrieval with source/line provenance. Separate telemetry validator.	Interoperability, harmonized event schema, data minimization and legal article/page metadata need extension. WCO/UN CEFACT formats are candidates to assess, not claims of local adoption.
M · Methods	Four role-specific retrieval/assessment capabilities; deterministic stage logic; optional LLM wording.	Contextualization, provenance, skills, evaluation and risk controls. No predictive agent is needed for the policy exercise; the separate classifier is experimental and

		synthetic.
P · Policies	Role-scoped disclosure; transfer hold; audit summary and human review gate.	Human oversight, accountability, contestation and institutional mandates. The prototype can flag missing legal context but cannot assign legal basis.
D · Distribution	Each assessment is joined into a stage trace with event IDs, citations, uncertainty and review state.	Human interface, language, accessibility, energy and deployment costs require user or operational measurement. No energy or inclusion metric is claimed.
SINK · Human users	Operators and competent officers are intended reviewers; all authority names remain generic/simulated.	Human-AI collaboration and domain capacity are design needs. Actual mandates, appeal path and accountable roles require stakeholder validation.

The ITU mapping distinguishes ARGOS application components from Y.3172's reference architecture. It does not claim ARGOS implements a future-network ML management system. AI Readiness dimensions such as contextualization, workflow integration, human interface, collaboration with AI, AI and policies, and energy/infrastructure are relevant, but only those with identified evidence are described as present.

5. Policy mapping and source quality

The current source register contains six entries: selected WTO Trade Facilitation Agreement provisions; selected, visually checked excerpts of Cameroon Law No. 2024/017; summaries/references for Cameroon railway law, the CEMAC merchant marine code, the OHADA road freight act, and the CEMAC Customs Code; plus a research inventory explicitly marked TO_VERIFY. A source-register status is metadata, not independent legal validation. The personal-data law excerpts are partial; implementing texts, factual applicability, and independent legal review remain outstanding.

Cameroon accepted the WTO Trade Facilitation Agreement Protocol on 30 November 2018, according to the WTO member page. The agreement is therefore a relevant international commitment, but the precise provision, Cameroon's implementation notifications, and the operational question must still be checked before drawing a conclusion. Articles 4, 7.4 and 11 are potential comparators for review/appeal, risk management and transit; they do not determine whether an ARGOS sensor event is admissible or who is responsible for a particular seal.

The official CNCC communiqué dated 15 June 2026 states that the requirement for a BESC for goods in transit is suspended until further notice. It says this followed a tripartite Cameroon–Chad–Central African Republic forum and refers to maintaining traceability. The communiqué itself does not establish that no replacement mechanism exists. The policy question for ARGOS is how to represent source freshness and unresolved process questions without inventing a rule.

6. Error scenarios and observed demonstration

Scenario A · Silent container

Expected behavior: roles receive only their allocated facts; the system does not infer a person's intent or assign fault; cited passages and uncertainty accompany assessments; late evidence can trigger reassessment; unsupported questions remain unresolved; a proposed location transfer waits for a human reviewer; and the audit output distinguishes fictional facts from legal sources.

Saved deterministic evidence: six stages (T0–T5), four simulated roles (A1–A4), 24 stage-level assessments/reconciliations, synthetic origin on every step, and source passages with source file and line metadata. At T3, transfer status is BLOCKED_PENDING_HUMAN_REVIEW; no personal data is transmitted and the human gate remains PENDING. The trace records lexical retrieval and explicitly states its limitations. Reproduction: PYTHONPATH=. .venv/bin/python scripts/run_policy_scenario.py --scenario silent_container_v1 --output /tmp/silent_container.json.

Scenario B · Profiling and data request

A fictional operator proposes using historical alerts to profile importers, while another party requests location histories. The desired response is to identify purpose, authority, recipient, retention and safeguards as unanswered; avoid treating historical alerts as proof of wrongdoing; cite what the corpus actually contains; identify the evidence gap; and escalate to a human. This is a proposed stress scenario, not an evaluated second replay. No discrimination metric or real customs workflow has been validated.

Retrieval control results

The saved control evaluation reports 12/12 expected source-presence/source-absence outcomes. It includes positive retrieval checks for bounded WTO and Cameroon-law excerpts and negative checks where national seal responsibility and the CNCC notice were not ingested in the tested corpus. This is a small retrieval-control set, not answer accuracy, legal correctness, or comprehensive retrieval performance.

7. Policy and standards gaps from the mapping

Gap / impact	Evidence and global reference	Next evidence needed
Location-data sharing: human gate exists, but operational purpose, lawful basis, recipients, retention and cross-border safeguards are not established by the prototype.	T3 holds the synthetic transfer. Partial Law 2024/017 excerpts are available. The law is a national source; WTO provisions are not a substitute for privacy law.	Complete primary law and implementing texts; map controller/processor roles and data flow; obtain independent legal review and define human approval/appeal procedure.
Seal and custody handovers: sensor events and checkpoint records can conflict, but software cannot determine custody or legal responsibility.	T4 deliberately supplies delayed location and broken-seal facts; national and regional transport/customs summaries are in the source inventory with applicability caveats.	Verify exact official customs, port, rail and carriage provisions and contracts; compare a global transit/seal practice using primary text; agree evidentiary protocol with operators.
BESC/transit process: official notice confirms suspension for transit goods until further notice; successor process and operational data exchange are not established here.	CNCC official communiqué, 15 June 2026. Do not infer absence of a replacement solely from this notice.	Obtain current CNCC/Customs and tripartite documents; verify process, responsible actors, event schema and dates. Mark unresolved elements as questions, not legal gaps.
Interoperability: prototype has its own event records; no official cross-system integration is demonstrated.	WCO Data Model and UN/CEFACT are candidates for comparison. No evidence here proves CAMCIS/SIGMAT deployment or incompatibility.	Acquire official interface specifications and adoption status; map identifiers, timestamps, seal/container references, quality and access rules.
AI recommendations and challenge: scenario proposes human review, but appeal, audit access and institutional accountability are not implemented.	Cameroon-law excerpts include provisions relevant to automated decisions/rights; excerpt scope and application need review. ITU AI Readiness includes policy, implementation and human-AI collaboration dimensions.	Confirm full legal text and responsible body; define review, challenge, correction and override paths; test bias and disparate impact before profiling use.
Evidence quality and field validity: synthetic scenarios cannot support real-corridor accuracy claims.	All policy replay data are synthetic. A separate logistic-regression artifact uses 301 synthetic samples across five scenarios; mean leave-one-scenario-out precision is 100%, recall 55.29% against synthetic labels only; it is not	Collect authorized field observations with calibration/provenance; predefine labels and route-separated evaluation; compare against rules and report errors.

loaded by operational alerts.

8. Conclusion and submission readiness

ARGOS demonstrates a bounded, replayable policy-coordination workflow: staged facts, role-specific context, source provenance, an explicit transfer hold and human review. Its strongest evidence is the deterministic scenario trace and the scoped source-control set. Main limitations are the small legal corpus, lexical retrieval, simulated institutional roles, synthetic operating events, and lack of independent legal/field evaluation.

The hackathon page specifies a five-page maximum technical report, a GitHub repository and a knowledge base of policies, regulations and strategies; a demo video is initially optional and required for shortlisted teams. Its published deadline is 10 November 2026. Team names/contact details and public repository access still need confirmation. This report is formatted to the five-page limit; verify final pagination after edits.

References and evidence inventory

- ITU, AI for Good Sandbox Hackathon: United Arab Emirates — criteria, materials and timeline. https://aiforgood.itu.int/event/ai-for-good-sandbox-hackathon-united-arab-emirates/ (accessed 7 Nov 2026).
- ITU-T Recommendation Y.3172, Architectural framework for machine learning in future networks including IMT-2020. https://www.itu.int/rec/T-REC-Y.3172/en
- ITU, AI Ready – Analysis Towards a Standardized Readiness Framework (2025). https://www.itu.int/epublications/publication/ai-ready-analysis-towards-a-standardized-readiness-framework
- Cameroon, Law No. 2024/017 of 23 December 2024, official law page. https://www.prc.cm/fr/actualites/actes/lois/7588 (local corpus contains selected excerpts only).
- WTO, Cameroon member information (TFA Protocol acceptance dated 30 Nov 2018). https://www.wto.org/english/thewto_e/countries_e/cameroon_e.htm
- WTO, Agreement on Trade Facilitation, official legal text. https://www.wto.org/english/docs_e/legal_e/tfa_e.htm
- CNCC, Communiqué on BESC requirement for transit goods, Douala, 15 June 2026. https://www.cncc.cm/fr/article/communique-1099
- African Union, Continental Artificial Intelligence Strategy (endorsed July 2024). https://au.int/en/documents/20240809/continental-artificial-intelligence-strategy
- Project evidence: knowledge_base/source_register.json; reports/argos_ai_for_good/evidence/silent_container_v1_deterministic.json; reports/argos_ai_for_good/evidence/policy_kb_retrieval_evaluation.json; reports/argos_ai_for_good/central_africa_model_validation_v2.json; app/agents/policy_scenario.py; docs/ITU_AI_READINESS_REFERENCE_INTEGRATION.md.

Evidence reviewed and report prepared 7 November 2026. Recheck the organizer page and primary references at submission. Working submission draft; team particulars and final organizational review remain to be supplied.